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Тема

Динамические аукционы с частными и общими компонентами ценности
при наличии инсайдерской информации

Title Dynamic Auctions with Insider Information and Private–Common Values

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Abstract

This paper studies a two-period second-price auction with asymmetric information and private–common values. One bidder, referred to as the insider, observes a persistent common-value component, while the uninformed bidder, referred to as the outsider, observes only its prior distribution. Both bidders also have independent private values in each period. Since the common component persists over time, the insider’s first-period bid affects not only the current allocation, but also the outsider’s posterior beliefs and hence the intensity of competition in the second period.

The model highlights an intertemporal trade-off faced by the informed bidder. A more aggressive first-period bid increases the probability of winning the current auction, but may reveal information about the common component and strengthen the outsider’s future bidding. Conversely, concealing information can reduce current payoff but soften future competition. We first analyze a class of monotone strategies in which the insider’s first-period bid depends only on the total valuation. This class allows for analytical characterization of beliefs and outsider best responses, but it is shown to be too restrictive.

In particular, the best response to truthful bidding, when the informed party bids his valuation of an object, contains pooling regions of positive measure that may suggest about this form of equilibrium behavior. We then develop a numerical best-response algorithm that allows the insider’s first-period strategy to depend separately on the common value and the private component. The algorithm constructs posterior beliefs from both continuous separating regions and atomic pooling regions, computes outsider best responses, and then evaluates the insider’s best response.

The numerical results confirm the analytical insight that pooling is a robust feature of the insider’s incentives. However, the best-response dynamics do not converge to a stable fixed point under the current implementation. The computations should therefore be interpreted as evidence on the structure of incentives rather than as a complete equilibrium characterization. They suggest that further work should focus on numerical methods explicitly adapted to semi-pooling equilibria and on a more refined treatment of beliefs after off-path deviations.

1 Introduction

Consider a procurement auction in which new firms compete against experienced suppliers for the execution of a contract. Experienced suppliers may possess informational advantages about the true cost or value of performing the contract. Beyond the terms explicitly stipulated in the contract, some relevant features of execution may remain unspecified but nevertheless affect all bidders. For example, in cleaning contracts, such a component may reflect the proportion of difficult rooms, such as laboratories, in the procurer’s building. This factor affects all suppliers’ costs but may be known only to the incumbent firm and the procurer (Albano et al., 2006).

In this environment, the inexperienced firm faces informational asymmetry regarding the value of the contract. However, this alone is not sufficient to describe the strategic conflict between the parties. To understand how firms interact under such asymmetry, one must account both for how the informed party uses its informational advantage and for how the uninformed party deals with this informational asymmetry. In this work, we therefore consider a dynamic environment in which the uninformed party, referred to as the outsider, deduces unknown part of the agreement by observing the actions, or the same their bids, of the informed party, referred to as the insider. Both parties are also heterogeneous in their private valuations of the contract, v_{kt} . For instance, firms may differ in productivity, and this productivity is privately known to each firm k given time t . Therefore, the outsider’s inference about the common value component, further denoted as c , which is insider’s private knowledge and it’s equally precious for both sides, is based both on its prior beliefs and on the observed actions of the insider. Anticipating this inference, the insider chooses its actions strategically.

The central question of this paper is how the informed bidder uses his informational advantage when current bids affect future beliefs. The insider would like to win the current auction when his valuation is high, but aggressive bidding may reveal information about the common component and make the outsider more competitive in the second period. The outsider, in turn, interprets the insider’s first-period bid as a signal and updates his beliefs about the common component accordingly. Thus, bids play a dual role: they determine the current allocation and simultaneously transmit information. The main contribution of this work is to combine the dynamic structure of the environment with bidders’ private–common values and show

that, together, these features can naturally model the core of the conflict and forecast the equilibrium of such interaction.

The first part of the analysis considers a restricted class of strategies in which the insider’s first-period bid is a strictly monotone function of the total valuation $w_{i1} = c + v_{i1}$. This restriction makes the belief structure tractable. Since the outsider can invert the insider’s bid, the posterior distribution of the common component is uniform over the set of feasible values consistent with the observed total valuation. Under this assumption, we derive the outsider’s sequentially rational strategies in both periods and express the insider’s payoff for a candidate strategy.

This monotone benchmark is useful for understanding the forces of the model, but it also reveals an important limitation. Linear monotone strategies are not sequentially rational, and even the best response to truthful bidding contains pooling regions. In particular, some sets of insider types with positive measure optimally choose the same first-period bid. This observation suggests that equilibrium behavior cannot generally be captured by strictly monotone strategies in the total valuation alone.

The second part of the paper therefore turns to a more general numerical approach. We allow the insider’s first-period strategy to be an arbitrary function $\beta_{i1}(c, v_{i1})$, so that it may contain both separating and pooling regions¹. The numerical algorithm constructs beliefs from the bid distribution induced by the insider’s strategy, taking into account both continuous likelihoods from locally separating regions and atomic likelihoods from pooling regions of positive measure value. Given these beliefs, the outsider’s strategies are computed by direct maximization, and the insider’s best response is evaluated over a finite bid grid.

The numerical results support the analytical finding that pooling incentives are central in this environment. Starting from truthful bidding, the best response computed by the algorithm reproduces the pooling regions predicted by the closed-form solution. Further iterations continue to generate strategies with substantial pooling. However, the best-response dynamics do not converge to a stable fixed point in the numerical experiments. Instead, the location and mass of pooling regions fluctuate across iterations. These results suggest that a candidate equilibrium, if it exists, is likely to be semi-pooling consisting of several regions positive measure where the insider pools his types, and that its computation requires methods specifically adapted to such structures.

2 Literature review

This work is related to the literature on auctions with asymmetric information, and in particular to models in which one bidder has superior information about a common component of value. A central issue in this literature is that bids serve not only as instruments for winning the object, but also as signals from which less informed bidders may learn.

A closely related paper is Hörner and Jamison (2008), who study sequential common-value auctions with asymmetrically informed bidders. In their model, one bidder has proprietary information about a persistent common value, while the uninformed bidder updates his beliefs from the history of observed bids and in this environment eventually the proprietary information is going to be fully disclosed by the insider. Our setting shares this learning channel: the informed side’s realized bid affects not only the current allocation, but also the outsider’s future beliefs. The main differences are that we allow bidders to have private-value components in addition to the common value, and we study a two-period environment rather than an infinite sequence of auctions.

Hörner and Jamison (2008) motivate their analysis by referring to wine auctions, where experts and non-experts ultimately value the same object but differ in their information about its quality. Experts observe quality more precisely, while non-experts rely on prior beliefs and on information revealed through bidding behavior. This interpretation is consistent with the empirical observations of Ashenfelter (1989), who document systematic patterns in wine-auction prices over time. Such patterns can be understood as reflecting learning by less informed bidders from earlier bidding behavior and drop-out decisions.

In addition, Virág (2003) obtains a related result in a repeated common-value auction setting without bid observation. Even when the uninformed bidder does not observe the informed bidder’s past bids, he may still update his beliefs about the object’s value by observing the outcomes of previous auctions. In

¹I call a strategy separating if each bid is chosen only by a measure-zero set of types, and pooling if bids are chosen by a positive-measure set of types.

equilibrium, more aggressive bidding can therefore serve an informational role, as it allows the uninformed bidder to obtain more precise information about the object’s value.

Information disclosure by insiders is also documented by Hernando-Veciana and Tröge (Hernando-Veciana and Tröge, 2011), who refer to this phenomenon as an insider’s curse, although the underlying mechanism differs. In contrast to Horner and Jamison’s work, they assume the presence of a private-value component that may be high for some outsiders, which becomes more likely when the number of bidders is large. As a result, given that players interact in a first price auction, such outsiders may value the object highly relative to most of their competitors and therefore bid more aggressively, a behavioral pattern observed experimentally (Kagel and Levin, 1999). Nonetheless, in contrast to the present work and other previously discussed studies, Hernando-Veciana and Tröge do not consider dynamic interaction among bidders and therefore does not analyze learning about the common value over time.

From the studies mentioned before we can see that the strategic behavior of outsiders and his beliefs formation play an important role in finding the equilibrium. This feature distinguishes the present paper from settings in which only the insider acts strategically, as in classic models of informed trading in financial markets (Kyle, 1985), where insider trader’s rent extraction from his private information is not affected by outsider’s updated beliefs.

A partially related to Hernando-Veciana and Tröge result is obtained by Bobkova (Bobkova, 2024), who shows that acquiring information about the common-value component makes bidders’ information more correlated and bidding behavior less independent across rivals, which may reduce aggressive bidding. At the same time, she finds that learning about the common component rather than one’s private component can lead to inefficiencies and exacerbate the winner’s curse.

Another important insight of from Bobkova is that when bidders are allowed to endogenously choose what information to acquire, certain auction formats and mechanism designs may induce bidders to learn about the common-value component instead of their private information. A related result is established by Bergemann et al. (Bergemann et al., 2009), who study English auctions with interdependent values and show that, in decentralized equilibrium, bidders tend to acquire costly information about their competitors’ signals more frequently than is socially efficient. In our work, we will not focus on the effect of mechanism design on the structure of equilibrium in our model and will use the second price auction for technical reasons. Nonetheless, taking into account previous literature, further analysis with other designs may be interesting to consider to check results robustness.

Model

Assumptions

The model considers two bidders, indexed by $k \in \{o, i\}$, competing over two time periods $t \in \{1, 2\}$. In each period, the players participate in a second-price auction for a single item. The valuation of bidder k in period t is given by

$$w_{kt} = v_{kt} + c,$$

where c is a common value component, identical for both bidders, and v_{kt} is a private value component specific to bidder k in period t .

Private values v_{kt} are independently drawn across bidders and periods from a uniform distribution:

$$v_{kt} \sim U[0, 1].$$

Each realization of v_{kt} is privately observed by bidder k in period t . Other players, as well as the same bidder with respect to future private values, know only the distribution.

Information about the common component c is asymmetric. At $t = 1$, the insider i observes the exact realization of c , whereas the outsider o only knows its prior distribution, which is assumed to be

$$c \sim U[0, 1].$$

c value is not affected by time, so the insider keeps its knowledge for the next period. At the same time, the outsider updates c distribution after observing b_{i1} , so we will denote c posterior measure as $\mu(c | b_{i1})$.

Another key assumption is that c is not learned by the outsider for $t = 2$ even if he wins the object at $t = 1$, so the outsider information about c distribution is affected only via observing b_{i1} value. This assumption allows us to consider a smaller variety of possible profitable deviations, specifically we will not consider the case when the outsider of a lower type v_{o1} will prefer to strategically bid more probably sacrificing his current utility to eradicate informational asymmetry in the next stage².

This assumption also implies that the outcome of the first-period auction does not alter the information structure in the second period. Consequently, the outsider's first-period optimization problem simplifies, since he does not need to account for the effect of the first-period winning status on future information. Therefore, in period 1 he maximizes his expected current utility rather than the total expected utility over the entire time horizon.

The timing of the game is as follows:

- **Period 1:** The outsider observes his private value v_{o1} , while the insider observes both v_{i1} and the common value c .
- Both bidders simultaneously submit bids b_{o1} and b_{i1} . The winner pays the second bid. Bids are publicly observed, and the outsider updates his beliefs $c|b_{i1}$ based on the insider's strategy.
- **Period 2:** The outsider observes v_{o2} , and the insider observes v_{i2} . The common value c remains unchanged.
- Both bidders submit bids b_{o2} and b_{i2} , and the winner pays the second bid.

The solution concept of the model is Perfect Bayesian Equilibrium. Formally, a Perfect Bayesian Equilibrium consists of a profile of strategies and a system of beliefs such that:

1. Each player's strategy is sequentially rational given his beliefs and the strategies of the other player;
2. Beliefs are updated according to Bayes' rule on the equilibrium path.

Regarding out-of-equilibrium beliefs, we presume that the outsider deduces that $c = 1$ almost surely if insider's deviation doesn't match to a set of actions given his strategy $\beta_{i1}(w_{i1})$, so the insider faces the most intense competition in the second period he may have in this game.

In the second period, the insider plays his weakly dominant strategy by bidding his valuation:

$$\beta_{i2}(v_{i2}, c) = v_{i2} + c.$$

In an equilibrium, it comprises a sequentially rational strategy for the insider in the second period. This assumption will allow us not to verify the first condition of a Bayesian equilibrium defined before as it holds automatically.

Search for an equilibrium

Given the assumptions we set before, we can introduce the approach of how search for an equilibrium may be done given some $\beta_{i1}(v_{i1}, c)$. Basically, given both insider's strategies $\beta_{i1}(v_{i1}, c)$ and $\beta_{i2}(v_{i1}, c) = v_{i1} + c$ we can derive sequentially rational outsider's strategies by solving the following maximization problems:

For the sequentially rational $\beta_{o2}(v_{o2}|b_{i1})$ with a consistent c posterior distribution $\mu(c | b_{i1})$ we should solve

$$U_{o2}(b_{o2}, \beta_{i2} | v_{o2}, b_{i1}) = E_{v_{i2}, \mu(c|b_{i1})} \left[(v_{o2} - v_{i2}) \mathbf{1}\{v_{i2} + c < b_{o2}\} \right] \xrightarrow{b_{o2}} \max$$

where $v_{i2} \sim U(0, 1)$, $c \sim \mu(c | b_{i1})$, v_{o2}, b_{i1} are given

Given that the outsider doesn't learn c value irrespective to an outcome of an auction, we need to solve the following problem to find sequentially rational $\beta_{o1}(v_{o1})$

²You can think about this assumption in the context of procurement auctions that both contracts are going to be executed after the suppliers will start working with them, so the outsider won't learn the exact value of c as he wins the first period auction given the strategy $\beta_{i1}(v_{i1}, c)$

$$U_{o1}(b_{o1}, \beta_{i1} \mid v_{o1}) = E_{v_{i1}, c} \left[(v_{o1} + c - \beta_{i1}(v_{i1}, c)) \mathbf{1}\{\beta_{i1}(v_{i1}, c) < b_{o1}\} \right] \xrightarrow{b_{o1}} \max$$

where $v_{i1}, c \sim U(0, 1)$, v_{o1} is given

Then, the only condition we need to verify to establish if the set of strategies $\beta_{o1}(v_{o1}), \beta_{o2}(v_{o2} \mid b_{i1}), \beta_{i1}(v_{i1}, c), \beta_{i2}(v_{i2}, c)$ and beliefs $\mu(c \mid b_{i1})$ comprise a Bayesian Nash equilibrium is whether the insider has a profitable deviation in the first period given outsider's strategies and beliefs from this set.

Strictly increasing $\beta_{i1}(w_{i1})$

Given a candidate first-period strategy $\beta_{i1}(v_{i1}, c)$ and the second-period strategy $\beta_{i2}(v_{i2}, c)$, one can characterize the outsider's sequentially rational strategies $\beta_{o1}(v_{o1})$ and $\beta_{o2}(v_{o2} \mid b_{i1})$, as well as a system of beliefs $\mu(c \mid b_{i1})$, using backward induction.

However, the insider's first-period strategy $\beta_{i1}(v_{i1}, c)$ is not known a priori and its finding constitutes a cornerstone of this work. The main difficulty arises from insider's intertemporal incentives to employ the strategy $\beta_{i1}(v_{i1}, c)$ which, on the one hand, should correspond well to the probability of winning an object of value $w_{i1} = v_{i1} + c$ in the first period with outsider's beliefs regarding c distribution, which in turn affects outsider's strategy in the second period $\beta_{o2}(v_{o2} \mid \beta_{i1}^{-1}(b_{i1}))$.

Let us illustrate this difficulty with a class of strictly increasing strategies on $w_{i1} = v_{i1} + c$, precisely

$$\beta_{i1}(v_{i1}, c) = \beta_{i1}(w_{i1}), \quad \text{with} \quad \frac{d\beta_{i1}(w_{i1})}{dw_{i1}} > 0.$$

In fact, such strategies assume that in equilibrium the insider treats alike all his types (c, v_{i1}) given that the object gives him the same utility w_{i1} , but this suggestion is irrelevant if the insider, particularly, mixes the realization of a high component c with types which correspond to lower realization of object's total value w_{i1} , mostly associated with low values of c . In particular, we may have

$$\frac{d\beta_{i1}(\bar{c}, \bar{v}_{i1})}{d\bar{c}} < \frac{d\beta_{i1}(\bar{c}, \bar{v}_{i1})}{d\bar{v}_{i1}}, \quad (\bar{c}, \bar{v}_{i1}) \in [0, 1]^2$$

So the insider's strategy in equilibrium is flatter on his type c , which allows him to disguise c with its wider range of possible realizations, thus increasing the range of $c \mid b_{i1}$ possible realizations.

Nonetheless, strictly increasing strategies $\beta_{i1}(w_{i1})$ allow us to calculate immediately the whole set of candidate equilibrium strategies and beliefs given a particular form of the strategy. In this work, given construction will not be used for finding Bayesian equilibrium of the game, but it may be helpful for guessing how a real equilibrium looks like.

Search with a strictly increasing insider's strategy Given that $\beta_{i1}(w_{i1})$ is strictly increasing in w_{i1} , the outsider can invert the insider's first-period bid and recover the corresponding total valuation

$$w_{i1} = \beta_{i1}^{-1}(b_{i1}).$$

Hence, after observing b_{i1} , the outsider knows that the pair (v_{i1}, c) must lie on the level set

$$v_{i1} + c = \beta_{i1}^{-1}(b_{i1}).$$

Because level sets of $v_{i1} + c$ are linear and (v_{i1}, c) is supported on the unit square, the posterior distribution of c conditional on b_{i1} is uniform on the interval of feasible values:

$$\mu(c \mid b_{i1}) \sim U(A_{\beta_{i1}}(b_{i1}), D_{\beta_{i1}}(b_{i1})),$$

where $A_{\beta_{i1}}(b_{i1})$ and $D_{\beta_{i1}}(b_{i1})$ denote, respectively, the lower and upper bounds of the support of the posterior distribution, which can be derived as:

$$A_{\beta_{i1}}(b_{i1}) = \max(0, \beta_{i1}^{-1}(b_{i1}) - 1), \quad D_{\beta_{i1}}(b_{i1}) = \min(1, \beta_{i1}^{-1}(b_{i1})),$$

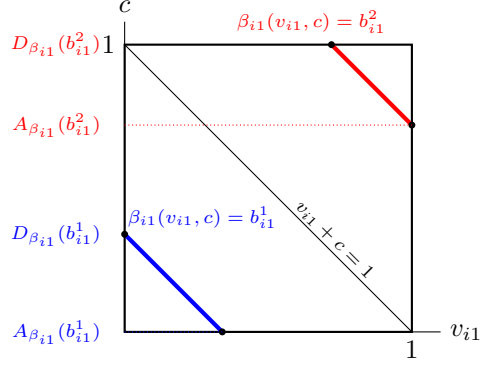


Figure 1: Illustration of outsider's belief formation under a strictly monotone strategy $\beta_{i1}(w_{i1})$ for $b_{i1}^1 < b_{i1}^2$.

We see that with strictly increasing strategies the insider pools values (c, v_{i1}) which in total give the same value $w_{i1} = v_{i1} + c$. This set of types (c, v_{i1}) is lying on a decreasing line with a slope of 45° . The higher is the insider's bid in the first period, the higher is level curve shifted, so boundaries $A_{\beta_{i1}}(b_{i1}), D_{\beta_{i1}}(b_{i1})$ move up correspondingly and for the outsider c ex post is distributed over a segment with its higher possible realizations.

Proposition 1. Given $\beta_{i1}(w_{i1})$, strategy of the outsider in the second period in a candidate equilibrium, which is derived in correspondence with the algorithm described before, constitutes a piecewise linear function of his type of the following form:

$$\beta_{o2}(v_{o2} \mid \beta_{i1}^{-1}(b_{i1})) = \begin{cases} A_{\beta_{i1}}(b_{i1}) + 2v_{o2}, & 0 \leq v_{o2} \leq \frac{L_{\beta_{i1}}(b_{i1})}{2}, \\ v_{o2} + \frac{\beta_{i1}^{-1}(b_{i1})}{2}, & \frac{L_{\beta_{i1}}(b_{i1})}{2} \leq v_{o2} \leq 1 - \frac{L_{\beta_{i1}}(b_{i1})}{2}, \\ D_{\beta_{i1}}(b_{i1}) - 1 + 2v_{o2}, & 1 - \frac{L_{\beta_{i1}}(b_{i1})}{2} \leq v_{o2} \leq 1. \end{cases}$$

$$A_{\beta_{i1}}(b_{i1}) = \max(0, \beta_{i1}^{-1}(b_{i1}) - 1), \quad D_{\beta_{i1}}(b_{i1}) = \min(1, \beta_{i1}^{-1}(b_{i1})), \\ L_{\beta_{i1}}(b_{i1}) = D_{\beta_{i1}}(b_{i1}) - A_{\beta_{i1}}(b_{i1}).$$

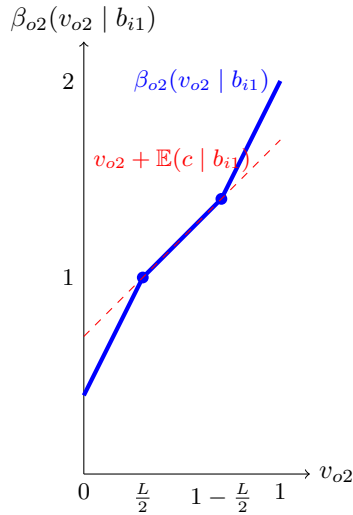


Figure 2: Illustration of the outsider's sequentially rational strategy in the second period

Proposition 2. Given $\beta_{i1}(w_{i1})$, strategy of the outsider in the first period in a candidate equilibrium, which is derived in correspondence with the algorithm described before, can be derived through the following functional equation:

$$\beta_{o1}(v_{o1}) - \frac{\beta_{i1}^{-1}(\beta_{o1}(v_{o1}))}{2} = v_{o1}$$

What's interesting with this first period strategy is that it shows that the outsider doesn't bid straightforwardly $\beta_{o2}(c, v_{o2}) = v_{o2} + E(c|b_{i1})$. It would happen so if both players possessed symmetric information regarding c value. In this case, the outsider is aware of the fact that the insider knows a particular value of c and bids correspondingly $c + v_{i2}$. As a result, ex post utility of the outsider from winning the object is $(v_{o2} - v_{i2})$, not $(v_{o2} + c - (v_{i2} - E(c|b_{i1})))$. Thus, given that the realization of v_{o2} is high, the outsider reaps ex post positive utility from winning the object if $v_{i2} < v_{o2}$, which is probable given that the probability of the realization of an opposite event is low; as a result the outsider has incentives to bid high. On the other hand, with a symmetric information positive utility is possible if $v_{o2} + c > v_{i2} + E(c|b_{i1})$, so ceteris paribus it's necessary to have a high realization of c and low v_{i2} , which is less probable and as a result disincentivizes the outsider to bid as high as he does under asymmetric information. The logic with low realization of v_{o2} is similar.

Regarding insider's second period strategy, here the exact form of $\beta_{i1}(w_{i1})$ is not specified, the only assumption regarding its form is based on its strict monotonicity by $w_{i1} = v_{i1} + c$. Assuming a particular form of $\beta_{i1}(w_{i1})$, form of $\beta_{o1}(v_{o1})$ can be derived consequently from the equation.

Given that the insider plays a strictly monotonous strategy $\beta_{i1}(w_{i1})$ in the first period and his weakly dominant strategy is to bid w_{i2} in the second period and that the outsider is sequentially rational and his beliefs are consistent, utility values in this candidate equilibrium are the following:

$$U_{i1}(\beta_{o1}, \beta_{i1} \mid v_{i1}, c) = w_{i1} \beta_{i1}(w_{i1}) - \frac{\beta_{i1}^2(w_{i1})}{2} - \frac{w_{i1}^2}{2} + \frac{1}{2} \int_0^{w_{i1}} \beta_{i1}(z) dz.$$

$$U_{i2}(\beta_{i1}, \beta_{o1}, \beta_{i2}, \beta_{o2} \mid c, b_{i1}) =$$

$$\frac{1}{6} - \frac{L_{\beta_{i1}}^3(b_{i1})}{48} + \frac{L_{\beta_{i1}}^2(b_{i1})}{8} + \frac{2 - L_{\beta_{i1}}(b_{i1})}{4} \left(c - \frac{\beta_{i1}^{-1}(b_{i1})}{2} \right)^2 + \frac{1}{2} \left(c - \frac{\beta_{i1}^{-1}(b_{i1})}{2} \right)$$

To find out if the equilibrium with strictly increasing $\beta_{i1}(w_{i1})$ exists, we can substitute $\beta_{i1}(w_{i1})$ with a specific function monotonous function to get the whole set of outsider's strategies and utility function values of the insider to see consequently if insider's best response in the first period differs from the equilibrium strategy $\beta_{i1}(w_{i1})$. In the next subsection we will look at linear on w_{i1} strategies to guess the equilibrium.

Linear $\beta_{i1}(w_{i1})$ strategy.

Assume that the insider plays a linear strategy in the first period:

$$\beta_{i1}(w_{i1}) = \alpha w_{i1}, \quad \alpha \in \left(\frac{1}{2}, 1 \right].$$

we will use this class of strategies as an illustration of how previous equations can be used to express particular forms of outsider's strategies and his beliefs.

Observing that $\beta_{i1}^{-1}(b) = b/\alpha$, the outsider's first-period best response is determined by

$$\beta_{o1}(v_{o1}) = \frac{2\alpha}{2\alpha - 1} v_{o1}.$$

The insider's candidate equilibrium first-period utility is

$$U_{i1}(\beta_{o1}, \beta_{i1} \mid v_{i1}, c) = \left(\frac{5\alpha}{4} - \frac{\alpha^2}{2} - \frac{1}{2} \right) w_{i1}^2, \quad w_{i1} = v_{i1} + c.$$

The insider's candidate equilibrium continuation utility in period 2 is

$$U_{i2}(\beta_{i1}, \beta_{o1}, \beta_{i2}, \beta_{o2} \mid c, \beta_{i1}(w_{i1})) = \frac{1}{6} - \frac{L(w_{i1})^3}{48} + \frac{L(w_{i1})^2}{8} + \frac{2 - L(w_{i1})}{4} \left(c - \frac{w_{i1}}{2} \right)^2 + \frac{1}{2} \left(c - \frac{w_{i1}}{2} \right),$$

where

$$L(w_{i1}) = \min\{w_{i1}, 2 - w_{i1}\}.$$

Therefore, for on-path deviations $b_{i1} \in [0, 2\alpha]$, the insider's total utility is

$$\begin{aligned}\hat{U}_i(b_{i1} | v_{i1}, c) &= (2\alpha - 1) \left(\frac{w_{i1}b_{i1}}{2\alpha} - \frac{b_{i1}^2}{4\alpha} \right) + \\ &+ \frac{1}{6} - \frac{L_\alpha(b_{i1})^3}{48} + \frac{L_\alpha(b_{i1})^2}{8} + \frac{2 - L_\alpha(b_{i1})}{4} \left(c - \frac{b_{i1}}{2\alpha} \right)^2 + \frac{1}{2} \left(c - \frac{b_{i1}}{2\alpha} \right).\end{aligned}$$

Lemma 1. Class of linear strategies $\beta_{i1}(w_{i1}) = \alpha w_{i1}$ is not sequentially rational. In particular, the insider finds profitable to deviate by bidding 0.8α instead of a prescribed action to bid α when $c = 1$ and $v_{i1} = 0$.

This lemma is indicative of the logic posited before with strictly increasing strategies, that the insider bids asymmetrically with his different types given object's value w_{i1} .

Pooling in Best Responses

In fact, profitable deviations arise not only for a class of linear strategies, but also for other multiple kinds of strictly monotonous $\beta_{i1}(w_{i1})$. The interesting part here is not the very fact of the existence of such deviations, but how the insider forms his best response, taking into account the strategies and belief structure of his uninformed opponent. In fact, there is evidence that an insider may have incentives to pool his types not only not monotonously, but also into sets with a non-zero measure by bidding correspondingly.

Specifically, from previous equations we know that if the insider plays in the first period a strategy $\beta_{i1}(w_{i1}) = w_{i1}$, outsider's reaction will be to use the following strategies

$$\begin{aligned}\beta_{o1}(v_{o1}) &= 2v_{o1} \\ \beta_{o2}(v_{o2} | b_{i1}) &= \begin{cases} A(b_{i1}) + 2v_{o2}, & 0 \leq v_{o2} \leq \frac{L_{\beta_{i1}}(b_{i1})}{2}, \\ v_{o2} + \frac{b_{i1}}{2}, & \frac{L(b_{i1})}{2} \leq v_{o2} \leq 1 - \frac{L(b_{i1})}{2}, \\ D(b_{i1}) - 1 + 2v_{o2}, & 1 - \frac{L(b_{i1})}{2} \leq v_{o2} \leq 1. \end{cases} \\ A(b_i) &= \max(0, b_{i1} - 1), D(b_{i1}) = \min(1, b_{i1}), L(b_{i1}) = D(b_{i1}) - A(b_{i1})\end{aligned}$$

And outsider's posterior of c is the following

$$\mu(c|b_{i1}) \sim U(A(b_{i1}), D(b_{i1}))$$

Also, after the substitution of $\beta_{i1}(w_{i1})$ with w_{i1} in insider's utility function values derived before, we can see that the outsider's best response pools types as it was mentioned before.

Lemma 2. Insider's best response $\hat{\beta}_{i1}(c, v_{i1})$ in the first period to outsider's strategies $\beta_{o1}(v_{o1}) = 2v_{o1}$, piecewise-linear $\beta_{o2}(v_{o2}|b_{i1})$ defined before with beliefs $\mu(c|b_{i1}) \sim U(A(b_{i1}), D(b_{i1}))$ has the following form:

$$\hat{\beta}_{i1}(c, v_{i1}) = \begin{cases} 0, & v_{i1} \leq \frac{1-c^2}{2}, \\ c - \sqrt{1-2v_{i1}}, & \frac{1-c^2}{2} < v_{i1} < \frac{1}{2}, \\ c + \sqrt{2v_{i1}-1}, & \frac{1}{2} \leq v_{i1} < \frac{1+(1-c)^2}{2}, \\ 1, & v_{i1} \geq \frac{1+(1-c)^2}{2}. \end{cases}$$

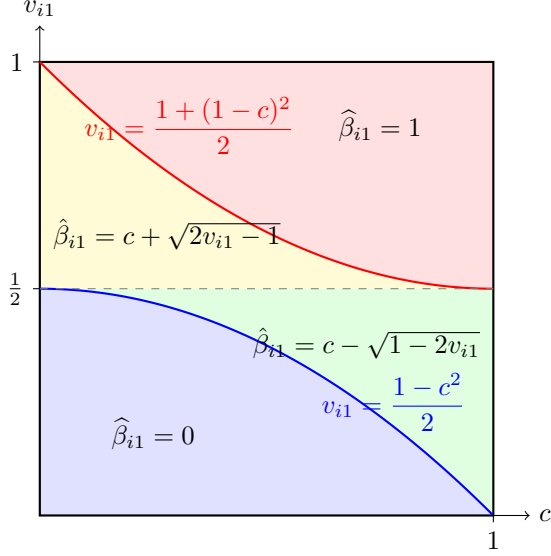


Figure 3: Regions of the insider's best response to outsider's strategies after truthful bidding.

In fact, this form of the best response suggests that, in the first period, the insider may have incentives to pool types by assigning the same bid to sets of positive measure. On the one hand, such pooling reduces the amount of information about c conveyed to the outsider before the second period. On the other hand, it can still preserve a reasonable first-period winning payoff. In the next section, we explore this pattern numerically by studying best-response dynamics and candidate equilibria of the model.

Search for an equilibrium, numerical methods

We saw previously that in an equilibrium the insider may prefer to play $\beta_{i1}(c, v_{i1})$ that pools his types in sets of non-zero measure, which is non-monotonic on w_{i1} by construction. Previous findings of how outsider's sequentially rational strategies and his beliefs were based on insider's monotonous strategy $\beta_{i1}(w_{i1})$, so now we need to consider a more general set-up to encompass a broader set of possible actions which may contain equilibrium strategies. For this purpose, we will employ numerical methods to formalize the algorithm of a search for an equilibrium in our model. we will present the algorithm in the next subsection.

Numerical Best-Response Algorithm: The numerical procedure is implemented on finite grids

$$\mathcal{C} = \{c_1, \dots, c_{n_c}\}, \quad \mathcal{V}_i = \{v_1^i, \dots, v_{n_i}^i\},$$

$$\mathcal{V}_o = \{v_1^o, \dots, v_{n_o}^o\}, \quad \mathcal{B} = \{b_1, \dots, b_{n_b}\}.$$

The insider's first-period strategy at iteration t is represented by a matrix

$$B_t^i[r, s] = \beta_{i1,t}(c_r, v_s^i).$$

Step 0. Initialization. Choose an initial insider strategy, for example

$$\beta_{i1,0}(c, v) = c + v.$$

If trembling hand is used for belief construction, define the belief-generating strategy

$$\beta_{i1,t}^{input}(c, v) = (1 - \varepsilon)\beta_{i1,t}(c, v) + \varepsilon(c + v).$$

If $\varepsilon = 0$, then

$$\beta_{i1,t}^{input}(c, v) = \beta_{i1,t}(c, v).$$

Step 1. Belief construction. For every possible first-period insider bid b , construct the posterior belief

$$\mu_t(c \mid b).$$

For a fixed c_r , consider the function

$$v_i \mapsto \beta_{i1,t}^{input}(c_r, v_i).$$

Between two adjacent grid points v_s^i and v_{s+1}^i , define

$$\Delta v = v_{s+1}^i - v_s^i,$$

and

$$\Delta b = \beta_{i1,t}^{input}(c_r, v_{s+1}^i) - \beta_{i1,t}^{input}(c_r, v_s^i).$$

If $\Delta b > tol_b$, the segment is treated as locally separating. It contributes the density

$$\frac{\Delta v}{|\Delta b|}$$

to all bid values lying between the two endpoint bids. This gives the continuous likelihood component

$$L_t^{cont}(b \mid c_r).$$

If $\Delta b < tol_b$, insider's strategy is considered to be flat on v_{i1} given b, c values. The mass

$$\Delta v$$

is assigned to the bid. In this case, conditioning on b_{i1} , the outsider deduces that c belongs to a region of types of non zero measure as we described with our example before. In this region, ex post probability of c is proportional to the mass of private components connected to this c value. So it gives the atomic likelihood component

$$L_t^{atom}(b^* \mid c_r) = \Delta v.$$

For each bid-grid point b_k , Bayes' rule is applied as follows. If b_k receives positive atomic mass for some c , the posterior is computed from the atomic likelihood:

$$\mu_t(c_r \mid b_k) = \frac{L_t^{atom}(b_k \mid c_r)}{\sum_{\ell=1}^{n_c} L_t^{atom}(b_k \mid c_\ell) \omega_\ell^c}.$$

If b_k does not receive atomic mass, the posterior is computed from the continuous likelihood:

$$\mu_t(c_r \mid b_k) = \frac{L_t^{cont}(b_k \mid c_r)}{\sum_{\ell=1}^{n_c} L_t^{cont}(b_k \mid c_\ell) \omega_\ell^c}.$$

Here ω_ℓ^c denotes the quadrature weight associated with c_ℓ . If a bid-grid point lies inside a numerical gap between two valid posterior rows, its posterior is filled by interpolation between the closest valid posterior distributions. If a bid lies outside the support generated by the insider's strategy, the off-path belief convention is used:

$$\mu_t(c = 1 \mid b) = 1.$$

Step 2. Outsider's second-period strategy. Given beliefs $\mu_t(c \mid b)$, the outsider's second-period strategy is computed by direct maximization. For every first-period bid b_k and outsider type v_m^o ,

$$\beta_{o2,t}(v_m^o \mid b_k) \in \arg \max_{q \in \mathcal{B}} \sum_{r=1}^{n_c} \sum_{s=1}^{n_i} (v_m^o - v_s^i) \mathbf{1}\{v_s^i + c_r < q\} \mu_t(c_r \mid b_k) \omega_r^c \omega_s^i.$$

Step 3. Outsider's first-period strategy. The outsider's first-period strategy is also computed by direct maximization. For each v_m^o ,

$$\beta_{o1,t}(v_m^o) \in \arg \max_{q \in \mathcal{B}} \sum_{r=1}^{n_c} \sum_{s=1}^{n_i} \left(v_m^o + c_r - \beta_{i1,t}^{input}(c_r, v_s^i) \right) \mathbf{1}\{ \beta_{i1,t}^{input}(c_r, v_s^i) < q \} \omega_r^c \omega_s^i.$$

Step 4. Insider's continuation value. For each common value c_r and each possible first-period bid b_k , compute the insider's expected second-period payoff:

$$U_{i2,t}(c_r \mid b_k) = \sum_{m=1}^{n_o} \sum_{s=1}^{n_i} \max \left\{ v_s^i + c_r - \beta_{o2,t}(v_m^o \mid b_k), 0 \right\} \omega_m^o \omega_s^i.$$

Step 5. Outsider first-period bid moments. Define

$$P_{o,t}(b_k) = \sum_{m=1}^{n_o} \mathbf{1}\{\beta_{o1,t}(v_m^o) < b_k\} \omega_m^o,$$

and

$$Q_{o,t}(b_k) = \sum_{m=1}^{n_o} \beta_{o1,t}(v_m^o) \mathbf{1}\{\beta_{o1,t}(v_m^o) < b_k\} \omega_m^o.$$

Then the insider's first-period expected payoff from bidding b_k is

$$(c_r + v_s^i)P_{o,t}(b_k) - Q_{o,t}(b_k).$$

Step 6. Insider's first-period best response. For every insider type (c_r, v_s^i) , compute

$$BR_t(c_r, v_s^i) \in \arg \max_{b_k \in \mathcal{B}} \left[(c_r + v_s^i)P_{o,t}(b_k) - Q_{o,t}(b_k) + U_{i2,t}(c_r \mid b_k) \right].$$

The next strategy is then defined by

$$\beta_{i1,t+1}(c_r, v_s^i) = BR_t(c_r, v_s^i).$$

Step 7. Diagnostics and stopping rule. At each iteration, we compute the distance between the current strategy and the induced best response. The main residual is defined as the L^2 distance

$$res_t = \left(\frac{1}{|\mathcal{C}||\mathcal{V}_i|} \sum_{c \in \mathcal{C}} \sum_{v_i \in \mathcal{V}_i} [\beta_{i1,t+1}(c, v_i) - \beta_{i1,t}(c, v_i)]^2 \right)^{1/2}.$$

A fixed point of the best-response procedure would require res_t to be close to zero. The looping stops when $res_t < tol_{alg}$

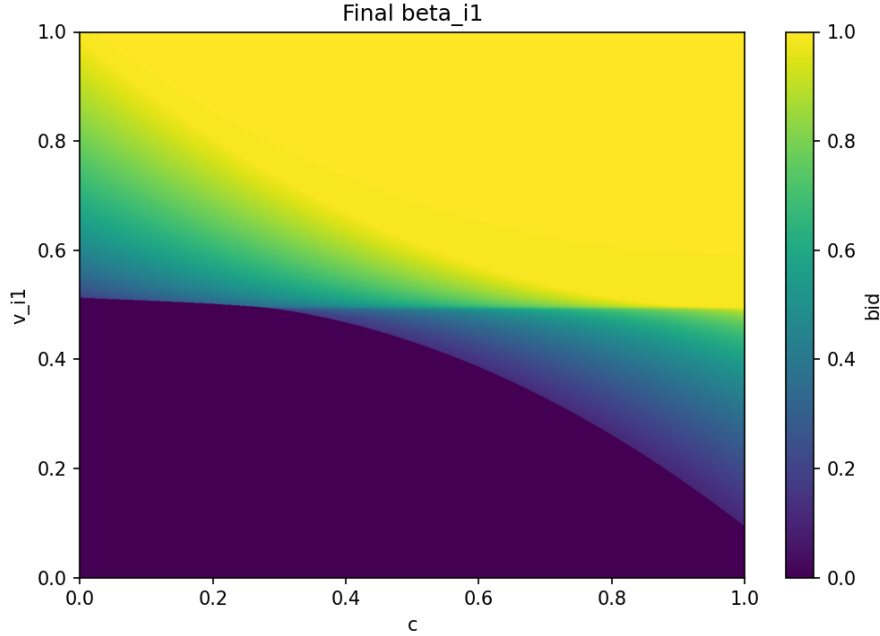


Figure 4: Regions of the insider's best response to outsider's strategies after truthful bidding, numerical implementation

Algorithm implementation. In this paper, we will report the results setting the following parameter values:

$$n_i = 1000, n_b = 200, \varepsilon = 0.03, tol_b = 0.1 * 10^{-10}, tol_{alg} = 0.1 * 10^{-4}$$

Figure 4 reports the numerical best response to the truthful first-period strategy $\beta_{i1}(w_{i1}) = w_{i1}$. The computed best response is consistent with the analytical solution derived above. In particular, the algorithm reproduces the two pooling regions, where the insider bids 0 and 1, as well as the continuous intermediate regions described by the closed-form best-response formula.

The results are qualitatively similar under other reasonable parameter choices. Nevertheless, it is worth noting that, for the grid sizes considered here, the residual res_t typically does not fall below $10^{-3} - 10^{-4}$, and in the following iteration the error rises appreciably. This indicates that the best-response dynamics do not converge to an exact fixed point within the numerical tolerance, even though the qualitative structure of the iterates remains stable across specifications. In the appendix we also provide algorithmic results with initialization with other insider's strategies.

Nonetheless, the numerical exercise therefore provides evidence that pooling may arise in equilibrium. After several iterations the insider's best response pools types on sets of positive measure. At the same time, as it was mentioned before, the numerical best-response procedure does not converge to a stable fixed point in the experiments. Iterating the algorithm from the truthful strategy produces strategies with pooling regions, but the location and mass of these regions continue to fluctuate over time. These best responses are depicted in the figures 5-10 below.

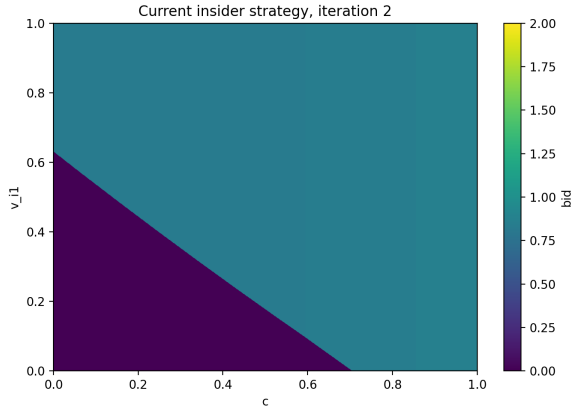


Figure 5: Best response, iteration 2

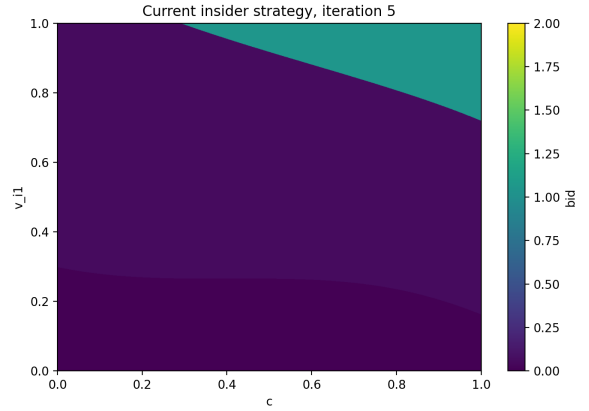


Figure 6: Best response, iteration 5

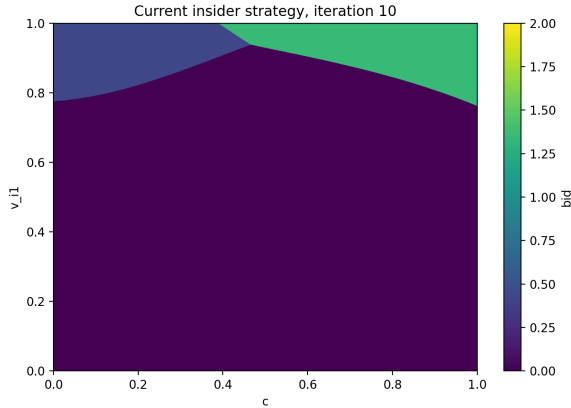


Figure 7: Best response, iteration 10

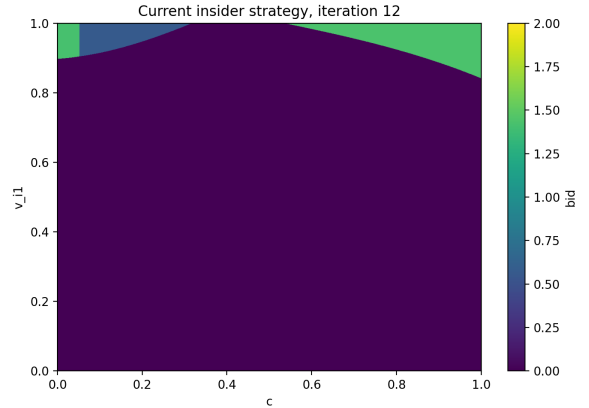


Figure 8: Best response, iteration 12

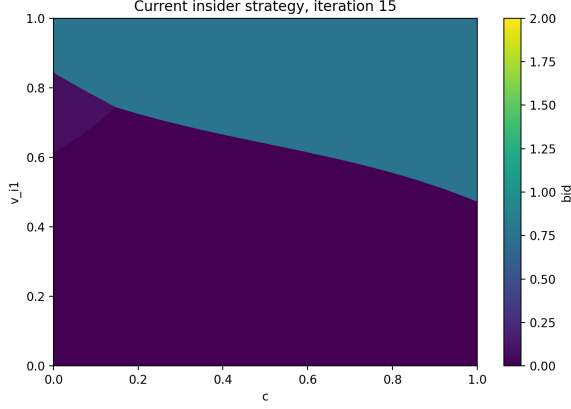


Figure 9: Best response, iteration 15

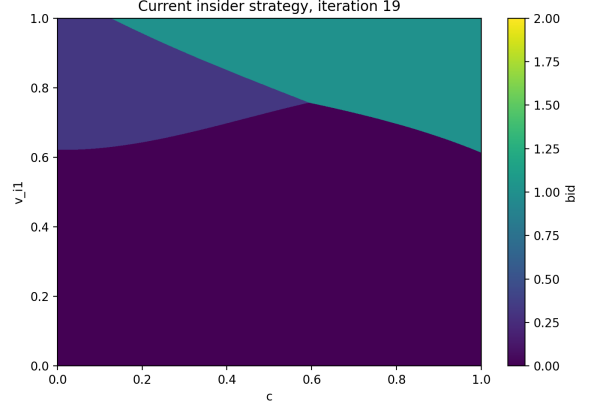


Figure 10: Best response, iteration 19

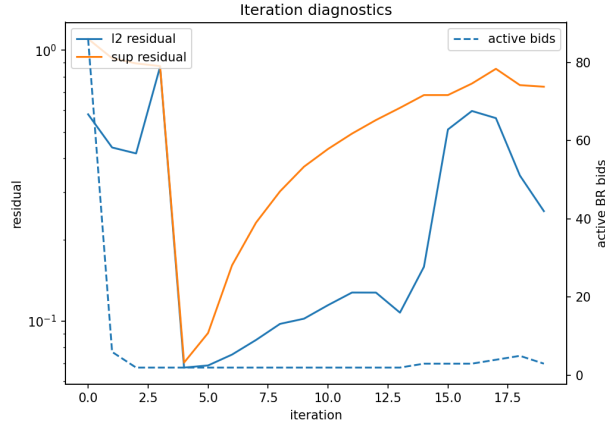


Figure 11: Iteration diagnostics starting from $\beta_{i1}(w_{i1}) = w_{i1}$. Number of active bids indicates the number of bids played with positive probability from insider's $BR_t(c, v_{i1})$

Hence, the computations should be interpreted as evidence about the structure of incentives rather than as a complete equilibrium computation. They suggest that a candidate equilibrium, if it exists, is likely to be semi-separating or partially pooling, but further work is required to characterize it precisely.

Further research should develop a numerical algorithm that is explicitly adapted to pooling behavior on the insider's side. The main difficulty appears to be related to off-equilibrium bids and the construction of beliefs after such deviations. Although trembling-hand perturbations help regularize beliefs, they do not fully eliminate the instability that arises when the insider's strategy contains pooling regions.

A promising direction is to refine the belief construction by allowing the posterior associated with a given bid to combine both continuous and atomic components. This would make the algorithm more suitable for strategies that are partially separating and partially pooling, and may help identify semi-separating equilibrium candidates more reliably.

Conclusion

This paper studies a two-period auction model in which one bidder has private information about a persistent common-value component. The model captures a dynamic strategic tension: the informed bidder can use his informational advantage in the current auction, but his first-period bid also affects the outsider's beliefs and therefore future competition. The outsider reacts to this asymmetry by treating the insider's bid as a signal about the common component and adjusting his subsequent bidding behavior.

The analysis begins with a tractable class of monotone first-period insider strategies depending only on the total valuation $w_{i1} = c + v_{i1}$. Under this restriction, the outsider’s posterior beliefs and sequentially rational strategies can be characterized analytically. This benchmark shows how the outsider infers information from the insider’s bid and how this inference affects second-period bidding. However, the monotone class turns out to be too restrictive. Linear monotone strategies are not sequentially rational, and the insider’s best response to truthful bidding already contains pooling regions of positive measure.

This finding is important because it indicates that the insider may optimally choose not only to shade his bid, but also to pool different types at the same first-period bid. Pooling reduces the amount of information transmitted about the common component and can therefore soften future competition. At the same time, it must still preserve sufficient current-period payoff. The closed-form best response to truthful bidding illustrates this trade-off directly: the insider pools low private-value types at bid zero and high private-value types at bid one, while using continuous bids only in intermediate regions.

To study richer strategies, we develop a numerical best-response algorithm that allows the insider’s first-period strategy to depend separately on c and v_{i1} . The algorithm constructs posterior beliefs from the induced bid distribution, allowing for both continuous separating regions and atoms generated by pooling. Given these beliefs, it computes the outsider’s sequentially rational strategies and then evaluates the insider’s best response. The numerical implementation reproduces the analytical best response to truthful bidding, providing a validation of the method in the benchmark case.

The numerical experiments suggest that pooling is a robust feature of the model. Across several initial strategies, the insider’s best response tends to generate regions in which many types choose the same bid. However, the best-response dynamics do not converge to a stable fixed point under the current implementation. Residuals remain bounded away from zero and the location of pooling regions fluctuates across iterations. Therefore, the numerical results should not be interpreted as a complete equilibrium computation. Rather, they provide evidence about the structure of incentives and suggest that any equilibrium candidate is likely to be semi-separating or partially pooling.

Several directions remain for future research. For instance, the numerical algorithm should be further adapted to pooling equilibria. In particular, belief construction after off-path deviations plays a crucial role, and trembling-hand perturbations do not fully resolve the instability generated by pooling regions. A promising extension is to allow posterior beliefs associated with a given bid to combine continuous and atomic components in a more systematic way.

Overall, the paper shows that dynamic information transmission can substantially alter bidding incentives in auctions with insider information. Even in a simple two-period second-price auction, the informed bidder’s best response may depart sharply from monotone bidding and involve pooling on sets of positive measure. This suggests that equilibrium analysis in such environments should allow for partially pooling strategies and should treat belief formation as a central object of the model.

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Appendix

Proof of the Proposition 1

The outsider's expected utility in period 2 can be written as

$$U_{o2}(b_{o2} \mid v_{o2}, b_{i1}) = v_{o2} \mathbb{P}(v_{i2} + c < b_{o2} \mid b_{i1}) - \mathbb{E}[v_{i2} \mathbf{1}\{v_{i2} + c < b_{o2}\} \mid b_{i1}],$$

and the outsider chooses b_{o2} to maximize this expression. We first compute the probability term. Conditional on a fixed value of c , we have

$$\mathbb{P}_{v_{i2}}(v_{i2} + c < b_{o2}) = \begin{cases} 0, & c \geq b_{o2}, \\ b_{o2} - c, & c \in (b_{o2} - 1, b_{o2}), \\ 1, & c \leq b_{o2} - 1, \end{cases}$$

since $v_{i2} \sim U[0, 1]$.

The unconditional probability is obtained by integrating over the posterior distribution

$$c \mid b_{i1} \sim U(A_{\beta_{i1}}(b_{i1}), D_{\beta_{i1}}(b_{i1})).$$

Let $L_{\beta_{i1}}(b_{i1}) = D_{\beta_{i1}}(b_{i1}) - A_{\beta_{i1}}(b_{i1})$ denote the length of this interval. Then

$$\mathbb{P}(v_{i2} + c < b_{o2} \mid b_{i1}) = \frac{1}{L_{\beta_{i1}}(b_{i1})} \int_{A_{\beta_{i1}}(b_{i1})}^{D_{\beta_{i1}}(b_{i1})} \mathbb{P}_{v_{i2}}(v_{i2} + c < b_{o2}) dc.$$

This expression can be written in piecewise form depending on the relative position of b_{o2} and the support of c :

$$\mathbb{P}(v_{i2} + c < b_{o2} \mid b_{i1}) = \begin{cases} 0, & A_{\beta_{i1}}(b_{i1}) \geq b_{o2}, \\ \frac{1}{L_{\beta_{i1}}(b_{i1})} \int_{A_{\beta_{i1}}(b_{i1})}^{b_{o2}} (b_{o2} - c) dc, & A_{\beta_{i1}}(b_{i1}) \leq b_{o2} \leq D_{\beta_{i1}}(b_{i1}), \\ \frac{1}{L_{\beta_{i1}}(b_{i1})} \int_{A_{\beta_{i1}}(b_{i1})}^{D_{\beta_{i1}}(b_{i1})} (b_{o2} - c) dc, & b_{o2} - 1 \leq A_{\beta_{i1}}(b_{i1}) \leq D_{\beta_{i1}}(b_{i1}) \leq b_{o2}, \\ \frac{1}{L_{\beta_{i1}}(b_{i1})} \left[\int_{A_{\beta_{i1}}(b_{i1})}^{b_{o2}-1} 1 dc + \int_{b_{o2}-1}^{D_{\beta_{i1}}(b_{i1})} (b_{o2} - c) dc \right], & A_{\beta_{i1}}(b_{i1}) \leq b_{o2} - 1 \leq D_{\beta_{i1}}(b_{i1}), \\ 1, & D_{\beta_{i1}}(b_{i1}) \leq b_{o2} - 1. \end{cases}$$

We now compute the second term in the outsider's objective function:

$$\mathbb{E}[v_{i2} \mathbf{1}\{v_{i2} + c < b_{o2}\} \mid b_{i1}].$$

Conditional on a fixed value of c , we have

$$\mathbb{E}_{v_{i2}}[v_{i2} \mathbf{1}\{v_{i2} + c < b_{o2}\}] = \int_0^1 v_{i2} \mathbf{1}\{v_{i2} < b_{o2} - c\} dv_{i2}.$$

Since $v_{i2} \sim U[0, 1]$, this becomes

$$\mathbb{E}_{v_{i2}}[v_{i2} \mathbf{1}\{v_{i2} + c < b_{o2}\}] = \begin{cases} 0, & c \geq b_{o2}, \\ \int_0^{b_{o2}-c} v_{i2} dv_{i2} = \frac{(b_{o2}-c)^2}{2}, & c \in (b_{o2} - 1, b_{o2}), \\ \int_0^1 v_{i2} dv_{i2} = \frac{1}{2}, & c \leq b_{o2} - 1. \end{cases}$$

Next, integrating over the posterior distribution

$$c \mid b_{i1} \sim U(A_{\beta_{i1}}(b_{i1}), D_{\beta_{i1}}(b_{i1})),$$

we obtain

$$\mathbb{E}[v_{i2}\mathbf{1}\{v_{i2} + c < b_{o2}\} \mid b_{i1}] = \frac{1}{L_{\beta_{i1}}(b_{i1})} \int_{A_{\beta_{i1}}(b_{i1})}^{D_{\beta_{i1}}(b_{i1})} \mathbb{E}_{v_{i2}}[v_{i2}\mathbf{1}\{v_{i2} + c < b_{o2}\}] dc,$$

where

$$L_{\beta_{i1}}(b_{i1}) = D_{\beta_{i1}}(b_{i1}) - A_{\beta_{i1}}(b_{i1}).$$

Therefore,

$$\mathbb{E}[v_{i2}\mathbf{1}\{v_{i2} + c < b_{o2}\} \mid b_{i1}] = \begin{cases} 0, & A_{\beta_{i1}}(b_{i1}) \geq b_{o2}, \\ \frac{1}{L_{\beta_{i1}}(b_{i1})} \int_{A_{\beta_{i1}}(b_{i1})}^{b_{o2}} \frac{(b_{o2} - c)^2}{2} dc, & A_{\beta_{i1}}(b_{i1}) \leq b_{o2} \leq D_{\beta_{i1}}(b_{i1}), \\ \frac{1}{L_{\beta_{i1}}(b_{i1})} \int_{A_{\beta_{i1}}(b_{i1})}^{D_{\beta_{i1}}(b_{i1})} \frac{(b_{o2} - c)^2}{2} dc, & b_{o2} - 1 \leq A_{\beta_{i1}}(b_{i1}) \leq D_{\beta_{i1}}(b_{i1}) \leq b_{o2}, \\ \frac{1}{L_{\beta_{i1}}(b_{i1})} \left[\int_{A_{\beta_{i1}}(b_{i1})}^{b_{o2}-1} \frac{1}{2} dc + \int_{b_{o2}-1}^{D_{\beta_{i1}}(b_{i1})} \frac{(b_{o2} - c)^2}{2} dc \right], & A_{\beta_{i1}}(b_{i1}) \leq b_{o2} - 1 \leq D_{\beta_{i1}}(b_{i1}), \\ \frac{1}{2}, & D_{\beta_{i1}}(b_{i1}) \leq b_{o2} - 1. \end{cases}$$

Given the previous expressions, we now derive the first-order condition by differentiating

$$U_{o2}(b_{o2} \mid v_{o2}, b_{i1})$$

with respect to b_{o2} .

Consider the case

$$A_{\beta_{i1}}(b_{i1}) \leq b_{o2} \leq D_{\beta_{i1}}(b_{i1}).$$

In this region, the outsider's expected utility is

$$U_{o2}(b_{o2} \mid v_{o2}, b_{i1}) = \frac{v_{o2}}{L_{\beta_{i1}}(b_{i1})} \int_{A_{\beta_{i1}}(b_{i1})}^{b_{o2}} (b_{o2} - c) dc - \frac{1}{L_{\beta_{i1}}(b_{i1})} \int_{A_{\beta_{i1}}(b_{i1})}^{b_{o2}} \frac{(b_{o2} - c)^2}{2} dc.$$

Applying the Leibniz rule, we obtain

$$\frac{\partial U_{o2}(b_{o2} \mid v_{o2}, b_{i1})}{\partial b_{o2}} = \frac{v_{o2}}{L_{\beta_{i1}}(b_{i1})} \int_{A_{\beta_{i1}}(b_{i1})}^{b_{o2}} 1 dc - \frac{1}{L_{\beta_{i1}}(b_{i1})} \int_{A_{\beta_{i1}}(b_{i1})}^{b_{o2}} (b_{o2} - c) dc.$$

Hence,

$$\frac{\partial U_{o2}(b_{o2} \mid v_{o2}, b_{i1})}{\partial b_{o2}} = \frac{1}{L_{\beta_{i1}}(b_{i1})} \left[v_{o2}(b_{o2} - A_{\beta_{i1}}(b_{i1})) - \frac{(b_{o2} - A_{\beta_{i1}}(b_{i1}))^2}{2} \right].$$

The first-order condition is therefore

$$v_{o2}(b_{o2} - A_{\beta_{i1}}(b_{i1})) - \frac{(b_{o2} - A_{\beta_{i1}}(b_{i1}))^2}{2} = 0.$$

Equivalently,

$$(b_{o2} - A_{\beta_{i1}}(b_{i1})) \left(v_{o2} - \frac{b_{o2} - A_{\beta_{i1}}(b_{i1})}{2} \right) = 0.$$

Then, the local maximum satisfying the constraints is

$$b_{o2}(v_{o2} \mid b_{i1}) = A_{\beta_{i1}}(b_{i1}) + 2v_{o2}$$

to meet the constraints, we should have

$$A_{\beta_{i1}}(b_{i1}) \leq A_{\beta_{i1}}(b_{i1}) + 2v_{o2} \leq D_{\beta_{i1}}(b_{i1})$$

which is equivalent to

$$0 \leq v_{o2} \leq \frac{L_{\beta_{i1}}(b_{i1})}{2}$$

Now let's consider the case when $b_{o2} - 1 \leq A \leq D \leq b_{o2}$. In this region, the outsider's expected utility is

$$U_{o2}(b_{o2} \mid v_{o2}, b_{i1}) = \frac{v_{o2}}{L_{\beta_{i1}}(b_{i1})} \int_{A_{\beta_{i1}}(b_{i1})}^{D_{\beta_{i1}}(b_{i1})} (b_{o2} - c) dc - \frac{1}{L_{\beta_{i1}}(b_{i1})} \int_{A_{\beta_{i1}}(b_{i1})}^{D_{\beta_{i1}}(b_{i1})} \frac{(b_{o2} - c)^2}{2} dc.$$

Differentiating with respect to b_{o2} , we obtain

$$\frac{\partial U_{o2}}{\partial b_{o2}} = v_{o2} - b_{o2} + \frac{A_{\beta_{i1}}(b_{i1}) + D_{\beta_{i1}}(b_{i1})}{2}.$$

The first-order condition is

$$v_{o2} - b_{o2} + \frac{A_{\beta_{i1}}(b_{i1}) + D_{\beta_{i1}}(b_{i1})}{2} = 0,$$

which implies

$$b_{o2} = v_{o2} + \frac{A_{\beta_{i1}}(b_{i1}) + D_{\beta_{i1}}(b_{i1})}{2}.$$

Then,

$$v_{o2} + \frac{A_{\beta_{i1}}(b_{i1}) + D_{\beta_{i1}}(b_{i1})}{2} - 1 \leq A_{\beta_{i1}}(b_{i1}) \leq D_{\beta_{i1}}(b_{i1}) \leq v_{o2} + \frac{A_{\beta_{i1}}(b_{i1}) + D_{\beta_{i1}}(b_{i1})}{2}$$

Which is equivalent to:

$$\frac{L_{\beta_{i1}}(b_{i1})}{2} \leq v_{o2} \leq 1 - \frac{L_{\beta_{i1}}(b_{i1})}{2}$$

Now consider $A \leq b_{o2} - 1 \leq D$. In this region, the utility is

$$\begin{aligned} U_{o2}(b_{o2} \mid v_{o2}, b_{i1}) &= \frac{v_{o2}}{L_{\beta_{i1}}(b_{i1})} \left[\int_{A_{\beta_{i1}}(b_{i1})}^{b_{o2}-1} 1 dc + \int_{b_{o2}-1}^{D_{\beta_{i1}}(b_{i1})} (b_{o2} - c) dc \right] \\ &\quad - \frac{1}{L_{\beta_{i1}}(b_{i1})} \left[\int_{A_{\beta_{i1}}(b_{i1})}^{b_{o2}-1} \frac{1}{2} dc + \int_{b_{o2}-1}^{D_{\beta_{i1}}(b_{i1})} \frac{(b_{o2} - c)^2}{2} dc \right]. \end{aligned}$$

Differentiating, we obtain

$$\frac{\partial U_{o2}}{\partial b_{o2}} = \frac{1}{L_{\beta_{i1}}(b_{i1})} \left[v_{o2}(D_{\beta_{i1}}(b_{i1}) - b_{o2} + 1) - \frac{1}{2} + \frac{(b_{o2} - D_{\beta_{i1}}(b_{i1}))^2}{2} \right].$$

The first-order condition is

$$v_{o2}(D_{\beta_{i1}}(b_{i1}) - b_{o2} + 1) - \frac{1}{2} + \frac{(b_{o2} - D_{\beta_{i1}}(b_{i1}))^2}{2} = 0.$$

Let $x = b_{o2} - D_{\beta_{i1}}(b_{i1})$. Then

$$x^2 - 2v_{o2}x + 2v_{o2} - 1 = 0,$$

with solutions

$$x = 1 \quad \text{or} \quad x = 2v_{o2} - 1.$$

Thus, the critical points are

$$b_{o2} = D_{\beta_{i1}}(b_{i1}) + 1 \quad \text{and} \quad b_{o2} = D_{\beta_{i1}}(b_{i1}) + 2v_{o2} - 1.$$

Therefore, the optimal bid in this region is

$$b_{o2} = D_{\beta_{i1}}(b_{i1}) + 2v_{o2} - 1.$$

Again,

$$A_{\beta_{i1}}(b_{i1}) \leq D_{\beta_{i1}}(b_{i1}) + 2v_{o2} - 2 \leq D_{\beta_{i1}}(b_{i1})$$

Which is equivalent to

$$1 - \frac{L_{\beta_{i1}}(b_{i1})}{2} \leq v_{o2} \leq 1$$

Putting the pieces together, the outsider's optimal second-period bid is

$$s_{o2}(v_{o2}, b) = \begin{cases} A_{\beta_{i1}}(b_{i1}) + 2v_{o2}, & 0 \leq v_{o2} \leq \frac{L_{\beta_{i1}}(b_{i1})}{2}, \\ v_{o2} + M_{\beta_{i1}}(b_{i1}), & \frac{L_{\beta_{i1}}(b_{i1})}{2} \leq v_{o2} \leq 1 - \frac{L_{\beta_{i1}}(b_{i1})}{2}, \\ D_{\beta_{i1}}(b_{i1}) - 1 + 2v_{o2}, & 1 - \frac{L_{\beta_{i1}}(b_{i1})}{2} \leq v_{o2} \leq 1. \end{cases}$$

Proof of the Proposition 2

Suppose the insider uses a strictly monotone first-period bidding rule

$$\beta_{i1}(v_{i1}, c) = \beta_{i1}(w_{i1}), \quad w_{i1} := v_{i1} + c \in [0, 2].$$

If the outsider submits bid b_{o1} , then he wins if and only if

$$b_{o1} > \beta_{i1}(w_{i1}) \iff w_{i1} < \beta_{i1}^{-1}(b_{o1}).$$

Since $w_{i1} = c + v_{i1}$ with $c, v_{i1} \sim U[0, 1]$, the density of w_{i1} is

$$f(w_{i1}) = \begin{cases} w_{i1}, & 0 \leq w_{i1} \leq 1, \\ 2 - w_{i1}, & 1 \leq w_{i1} \leq 2. \end{cases}$$

Conditional on w_{i1} , we have

$$E[c \mid w_{i1}] = E[w_{i1} - v_{i1} \mid w_{i1}] = w_{i1} - \frac{w_{i1}}{2} = \frac{w_{i1}}{2}.$$

Hence the outsider's expected first-period utility from bid b_{o1} is

$$U(b_{o1} \mid v_{o1}) = \int_0^{\beta_{i1}^{-1}(b_{o1})} \left(v_{o1} + \frac{w_{i1}}{2} - \beta_{i1}(w_{i1}) \right) f(w_{i1}) dw_{i1}.$$

Let $t = \beta_{i1}^{-1}(b_{o1})$. Then the outsider chooses $t \in [0, 2]$ to maximize

$$\int_0^t \left(v_{o1} + \frac{w_{i1}}{2} - \beta_{i1}(w_{i1}) \right) f(w_{i1}) dw_{i1}.$$

Therefore an interior optimum satisfies

$$v_{o1} + \frac{t}{2} - \beta_{i1}(t) = 0,$$

or equivalently

$$\beta_{i1}(\beta_{i1}^{-1}(b_{o1})) - \frac{\beta_{i1}^{-1}(b_{o1})}{2} = v_{o1}$$

Then, the best response $\beta_{o1}(v_{o1})$ should meet the following condition:

$$\beta_{o1}(v_{o1}) - \frac{\beta_{i1}^{-1}(\beta_{o1}(v_{o1}))}{2} = v_{o1}$$

Regarding the second order condition:

$$\frac{1}{2} - \beta'_{i1}(t) < 0, \forall t = \beta_{i1}^{-1}(\beta_{o1}(v_{o1}))$$

Derivation of insider's utility function value in a candidate equilibrium with given $\beta_{i1}(w_{i1})$

Insider's first period ex ante utility

$$E_{v_{o1}} \left((w_{i1} - \beta_{o1}(v_{o1})) I(\beta_{o1}(v_{o1}) \leq \hat{b}) \right) = \int_0^1 (w_{i1} - \beta_{o1}(v_{o1})) I(\beta_{o1}(v_{o1}) \leq \hat{b}) dv_{o1}.$$

We can again substitute $\beta_{i1}^{-1}(\beta_{o1}(v_{o1})) = w_{i1}$ and knowing the relationship between $\beta_{i1}(w_{i1})$ and $\beta_{o1}(v_{o1})$

$$\beta_{o1}(v_{o1}) - \frac{\beta_{i1}^{-1}(\beta_{o1}(v_{o1}))}{2} = v_{o1},$$

get the following substitution

$$\beta_{i1}(w_{i1}) - \frac{w_{i1}}{2} = v_{o1}$$

So as before we get

$$\hat{U}_{i1} = \int_0^{\hat{w}_{i1}} (w_{i1} - \beta_{i1}(z)) (\beta'_{i1}(z) - \frac{1}{2}) dz$$

where $\hat{w}_{i1} = \beta_{i1}^{-1}(\hat{b})$. We can notice that

$$\int_0^{\hat{w}_{i1}} \beta'_{i1}(z) \beta_{i1}(z) dz = \int_0^{\hat{w}_{i1}} \beta_{i1}(z) d\beta_{i1}(z) = \frac{\beta_{i1}^2(z)}{2} \Big|_0^{\hat{w}_{i1}} = \frac{\beta_{i1}^2(\hat{w}_{i1})}{2}$$

Knowing that, we can express insider's first period utility function value when he bids $\hat{b}$ and the outsider acts according to the algorithm defined before has the following form:

$$\hat{U}_{i1} = w_{i1} \hat{b} - \frac{\hat{b}^2}{2} - \frac{w}{2} \hat{w}_{i1} + \frac{1}{2} \int_0^{\hat{w}_{i1}} \beta_{i1}(z) dz$$

In a candidate equilibrium, when $\hat{b} = \beta_{i1}(w_{i1})$ and $\hat{w}_{i1} = w_{i1}$, we have

$$U_{i1}(\beta_{o1}, \beta_{i1} \mid v_{i1}, c) = w_{i1} \beta_{i1}(w_{i1}) - \frac{\beta_{i1}^2(w_{i1})}{2} - \frac{w_{i1}^2}{2} + \frac{1}{2} \int_0^{w_{i1}} \beta_{i1}(z) dz.$$

Insider's continuation utility in the second period

Fix $b_{i1} \in [0, 2]$ and $c \in [A_{\beta_{i1}}(b_{i1}), D_{\beta_{i1}}(b_{i1})]$. The insider's continuation utility is

$$U_{i2}(c \mid b_{i1}) = E_{v_{o2}, v_{i2}} \left[(v_{i2} + c - \beta_{o2}(v_{o2} \mid b_{i1})) \mathbf{1}\{\beta_{o2}(v_{o2} \mid b_{i1}) < v_{i2} + c\} \right].$$

To simplify notation, write

$$A := A_{\beta_{i1}}(b_{i1}), \quad D := D_{\beta_{i1}}(b_{i1}), \quad L := D - A, \quad M := \frac{A + D}{2}.$$

Notice that

$$A = M - \frac{L}{2}, \quad D = M + \frac{L}{2}, \quad c \in [A, D].$$

It is convenient to define

$$u := c - M.$$

Then

$$u \in \left[-\frac{L}{2}, \frac{L}{2} \right].$$

The outsider's second-period best response is

$$\beta_{o2}(v_{o2} \mid b_{i1}) = \begin{cases} A + 2v_{o2}, & 0 \leq v_{o2} \leq \frac{L}{2}, \\ v_{o2} + M, & \frac{L}{2} \leq v_{o2} \leq 1 - \frac{L}{2}, \\ D - 1 + 2v_{o2}, & 1 - \frac{L}{2} \leq v_{o2} \leq 1. \end{cases}$$

We first integrate with respect to v_{i2} . Fix v_{o2} , and denote

$$s := \beta_{o2}(v_{o2} \mid b_{i1}).$$

Then

$$E_{v_{i2}} \left[(v_{i2} + c - s) \mathbf{1}\{s < v_{i2} + c\} \right] = \int_0^1 (v_{i2} + c - s) \mathbf{1}\{v_{i2} > s - c\} dv_{i2}.$$

Since $v_{i2} \sim U[0, 1]$, the correct lower bound is $\max\{0, s - c\}$, so

$$E_{v_{i2}} \left[(v_{i2} + c - s) \mathbf{1}\{s < v_{i2} + c\} \right] = \int_{\max\{0, s - c\}}^1 (v_{i2} + c - s) dv_{i2}.$$

Therefore

$$E_{v_{i2}} \left[(v_{i2} + c - s) \mathbf{1}\{s < v_{i2} + c\} \right] = \begin{cases} \frac{1}{2} + c - s, & s \leq c, \\ \frac{(1+c-s)^2}{2}, & c < s < c + 1, \\ 0, & s \geq c + 1. \end{cases}$$

We now integrate over v_{o2} , splitting into the three regions of β_{o2} .

First region. For $0 \leq v_{o2} \leq \frac{L}{2}$, we have

$$s = A + 2v_{o2} = M - \frac{L}{2} + 2v_{o2}.$$

The condition $s \leq c$ becomes

$$M - \frac{L}{2} + 2v_{o2} \leq M + u \iff v_{o2} \leq \frac{u + L/2}{2}.$$

Hence

$$I_1 = \int_0^{(u+L/2)/2} \left(\frac{1}{2} + c - (A + 2v) \right) dv + \int_{(u+L/2)/2}^{L/2} \frac{(1 + c - (A + 2v))^2}{2} dv.$$

Substituting $c = M + u$ and $A = M - \frac{L}{2}$,

$$I_1 = \int_0^{(u+L/2)/2} \left(\frac{1}{2} + u + \frac{L}{2} - 2v \right) dv + \int_{(u+L/2)/2}^{L/2} \frac{(1 + u + \frac{L}{2} - 2v)^2}{2} dv.$$

Direct calculation yields

$$I_1 = \frac{L^3}{96} - \frac{L^2 u}{16} + \frac{L u^2}{8} + \frac{L u}{2} + \frac{L}{4} - \frac{u^3}{12}.$$

Middle region. For $\frac{L}{2} \leq v_{o2} \leq 1 - \frac{L}{2}$, we have

$$s = v_{o2} + M.$$

Since $v_{o2} \geq \frac{L}{2} \geq u$, we have $s \geq c$. Also $s \leq c + 1$ on this interval. Therefore only the quadratic case applies:

$$I_2 = \int_{L/2}^{1-L/2} \frac{(1 + c - (M + v))^2}{2} dv = \int_{L/2}^{1-L/2} \frac{(1 + u - v)^2}{2} dv.$$

Direct calculation gives

$$I_2 = -\frac{L^3}{24} + \frac{L^2}{8} - \frac{L u^2}{2} - \frac{L u}{2} - \frac{L}{4} + \frac{u^2}{2} + \frac{u}{2} + \frac{1}{6}.$$

Third region. For $1 - \frac{L}{2} \leq v_{o2} \leq 1$, we have

$$s = D - 1 + 2v_{o2} = M + \frac{L}{2} - 1 + 2v_{o2}.$$

Here $s \geq c$, and the condition $s < c + 1$ becomes

$$M + \frac{L}{2} - 1 + 2v_{o2} < M + u + 1 \iff v_{o2} < 1 + \frac{u}{2} - \frac{L}{4}.$$

Hence

$$I_3 = \int_{1-L/2}^{1+u/2-L/4} \frac{(1 + c - (D - 1 + 2v))^2}{2} dv.$$

Substituting $c = M + u$ and $D = M + \frac{L}{2}$,

$$I_3 = \int_{1-L/2}^{1+u/2-L/4} \frac{(2 + u - \frac{L}{2} - 2v)^2}{2} dv.$$

Direct calculation yields

$$I_3 = \frac{L^3}{96} + \frac{L^2 u}{16} + \frac{L u^2}{8} + \frac{u^3}{12}.$$

Summing the three regions. Adding I_1 , I_2 , and I_3 , we obtain

$$I_1 + I_2 + I_3 = \frac{1}{6} - \frac{L^3}{48} + \frac{L^2}{8} + \frac{2-L}{4}u^2 + \frac{u}{2}.$$

Therefore

$$U_{i2}(c \mid b_{i1}) = I_1 + I_2 + I_3 = \frac{1}{6} - \frac{L^3}{48} + \frac{L^2}{8} + \frac{2-L}{4}u^2 + \frac{u}{2}.$$

Recalling that $u = c - M(b_{i1})$, we conclude that

$$\begin{aligned} & \boxed{U_{i2}(\beta_{i1}, \beta_{o1}, \beta_{i2}, \beta_{o2} \mid c, b_{i1}) =} \\ & = \boxed{\frac{1}{6} - \frac{L(b_{i1})^3}{48} + \frac{L(b_{i1})^2}{8} + \frac{2-L(b_{i1})}{4}(c - M(b_{i1}))^2 + \frac{1}{2}(c - M(b_{i1}))}, \\ & \boxed{c \in [A(b_{i1}), D(b_{i1})]} \end{aligned}$$

Technical part: extension of $U_{i2}(c \mid b_{i1})$ to off-path types

The expression for $U_{i2}(c \mid b_{i1})$ derived in the previous subsection is valid only for $c \in [A_{\beta_{i1}}(b_{i1}), D_{\beta_{i1}}(b_{i1})]$, i.e., when the true common value lies inside the outsider's posterior support. However, when evaluating potential deviations of the insider, the bid b_{i1} may be chosen in such a way that the corresponding posterior $[A(b_{i1}), D(b_{i1})]$ does not contain the insider's true c . We therefore extend the derivation to allow for arbitrary $c \in [0, 1]$.

To lighten notation, in what follows we write

$$A := A_{\beta_{i1}}(b_{i1}), \quad D := D_{\beta_{i1}}(b_{i1}), \quad L := D - A, \quad M := \frac{A+D}{2}.$$

Recall that

$$U_{i2}(c \mid b_{i1}) = E_{v_{o2}, v_{i2}} \left[(v_{i2} + c - \beta_{o2}(v_{o2} \mid b_{i1})) \mathbf{1}\{\beta_{o2}(v_{o2} \mid b_{i1}) < v_{i2} + c\} \right],$$

and that the outsider's second-period best response is piecewise linear:

$$\beta_{o2}(v_{o2} \mid b_{i1}) = \begin{cases} A + 2v_{o2}, & 0 \leq v_{o2} \leq \frac{L}{2}, \\ v_{o2} + M, & \frac{L}{2} \leq v_{o2} \leq 1 - \frac{L}{2}, \\ D - 1 + 2v_{o2}, & 1 - \frac{L}{2} \leq v_{o2} \leq 1. \end{cases}$$

For fixed v_{o2} , denote $s := \beta_{o2}(v_{o2} \mid b_{i1})$. Integrating first with respect to $v_{i2} \sim U[0, 1]$ yields

$$E_{v_{i2}} \left[(v_{i2} + c - s) \mathbf{1}\{v_{i2} + c > s\} \right] = \begin{cases} \frac{1}{2} + c - s, & s \leq c, \\ \frac{(1+c-s)^2}{2}, & c < s < c + 1, \\ 0, & s \geq c + 1. \end{cases}$$

We refer to these three cases as the *linear*, *quadratic*, and *zero* cases. Writing

$$U_{i2}(c \mid b_{i1}) = I_1 + I_2 + I_3,$$

we now recompute I_1 , I_2 , I_3 for arbitrary $c \in [0, 1]$.

Region 1: $v_{o2} \in [0, L/2]$, $s \in [A, D]$. As v_{o2} ranges over $[0, L/2]$, s sweeps the posterior support $[A, D]$. Which of the three cases applies depends on the position of c relative to $[A, D]$:

- If $c \geq D$, then $s \leq D \leq c$ throughout, and the linear case applies on the entire region;
- If $A \leq c \leq D$, then the linear case applies for $s \in [A, c]$ and the quadratic case for $s \in [c, D]$, with the split at $v_{o2} = (c - A)/2$;
- If $c < A$, then $s > c$ throughout and the quadratic case applies on the entire region. (The zero case would require $s \geq c + 1$, which would need $A \geq c + 1$, impossible for $c \geq 0$ and $A \leq 1$.)

Direct integration yields

$$I_1 = \begin{cases} \int_0^{L/2} \left(\frac{1}{2} + c - A - 2v\right) dv, & c \geq D, \\ \int_0^{(c-A)/2} \left(\frac{1}{2} + c - A - 2v\right) dv + \int_{(c-A)/2}^{L/2} \frac{(1 + c - A - 2v)^2}{2} dv, & A \leq c \leq D, \\ \int_0^{L/2} \frac{(1 + c - A - 2v)^2}{2} dv, & c \leq A. \end{cases}$$

Region 2: $v_{o2} \in [L/2, 1 - L/2]$, $s \in [D, 1 + A]$. Here s sweeps the interval $[D, 1 + A]$. The relevant kinks of the inner function are at $s = c$ and $s = c + 1$:

- The kink $s = c$ lies inside $[D, 1 + A]$ iff $D \leq c \leq 1 + A$. Since $c \leq 1 \leq 1 + A$, this reduces to $c \geq D$;
- The kink $s = c + 1$ lies inside $[D, 1 + A]$ iff $D - 1 \leq c \leq A$. Since $c \geq 0 \geq D - 1$ for $D \leq 1$, this reduces to $c \leq A$.

Hence:

- If $c \geq D$, the linear case applies for $s \in [D, c]$ and the quadratic case for $s \in [c, 1 + A]$, with the split at $v_{o2} = c - M$;
- If $A \leq c \leq D$, only the quadratic case applies (this is the on-path case);
- If $c \leq A$, the quadratic case applies for $s \in [D, c + 1]$ and the zero case for $s \in [c + 1, 1 + A]$, with the split at $v_{o2} = c + 1 - M$.

Therefore,

$$I_2 = \begin{cases} \int_{L/2}^{c-M} \left(\frac{1}{2} + c - v - M\right) dv + \int_{c-M}^{1-L/2} \frac{(1 + c - v - M)^2}{2} dv, & c \geq D, \\ \int_{L/2}^{1-L/2} \frac{(1 + c - v - M)^2}{2} dv, & A \leq c \leq D, \\ \int_{L/2}^{c+1-M} \frac{(1 + c - v - M)^2}{2} dv, & c \leq A. \end{cases}$$

Region 3: $v_{o2} \in [1 - L/2, 1]$, $s \in [1 + A, D + 1]$. Here s sweeps the interval $[1 + A, D + 1]$. The kink $s = c$ lies in this interval iff $1 + A \leq c \leq D + 1$, which for $c \in [0, 1]$ holds only in the degenerate case $A = 0$, $c = 1$. The kink $s = c + 1$ lies in $[1 + A, D + 1]$ iff $A \leq c \leq D$. Hence:

- If $c \geq D$, then $s \geq 1 + A \geq c - L + 1 \geq c - 1 + 1 = c$... more precisely, at $v_{o2} = 1 - L/2$ we have $s = 1 + A$ and $c - s = c - 1 - A \in [-A, -A + 1 - D] \subseteq [-1, 0]$, so we start in the quadratic case; the zero-case kink $v_{o2} = 1 + (c - D)/2$ lies above 1, hence outside the region. The quadratic case applies throughout;
- If $A \leq c \leq D$, the quadratic case applies for $v_{o2} \in [1 - L/2, 1 + (c - D)/2]$ and the zero case beyond (this is the on-path case);
- If $c \leq A$, then at $v_{o2} = 1 - L/2$, $c - s = c - 1 - A \leq A - 1 - A = -1$, so the zero case applies throughout and $I_3 = 0$.

Therefore,

$$I_3 = \begin{cases} \int_{1-L/2}^1 \frac{(1 + c - (D - 1 + 2v))^2}{2} dv, & c \geq D, \\ \int_{1-L/2}^{1+(c-D)/2} \frac{(1 + c - (D - 1 + 2v))^2}{2} dv, & A \leq c \leq D, \\ 0, & c \leq A. \end{cases}$$

Assembling the three regions. Adding the three terms in each case and simplifying algebraically, one obtains a closed-form expression for $U_{i2}(c \mid b_{i1})$ valid on all of $[0, 1]$. It is convenient to write the result as the on-path formula plus a correction term:

$$U_{i2}(c \mid b_{i1}) = U_{i2}^{\text{on}}(c \mid b_{i1}) + \Delta(c \mid b_{i1}),$$

where

$$U_{i2}^{\text{on}}(c \mid b_{i1}) = \frac{1}{6} - \frac{L^3}{48} + \frac{L^2}{8} + \frac{2-L}{4}(c-M)^2 + \frac{1}{2}(c-M)$$

is the on-path expression derived previously, and

$$\Delta(c \mid b_{i1}) = \begin{cases} -\frac{(c-D)^3}{6}, & c \geq D, \\ 0, & A \leq c \leq D, \\ +\frac{(c-A)^3}{6}, & c \leq A. \end{cases}$$

Proof of the lemma 1

Now consider the type $c = 1$, $v_{i1} = 0$, so that $w_{i1} = 1$. Under the candidate linear strategy, the insider bids

$$b_{i1}^{\text{cand}} = \alpha.$$

Consider instead the deviation to the bid corresponding to the type $w_{i1} = 0.8$, namely

$$b_{i1}^{\text{dev}} = 0.8\alpha.$$

The candidate total utility is

$$U_i^{\text{cand}}(\alpha \mid 0, 1) = \frac{5\alpha}{4} - \frac{\alpha^2}{2} + \frac{1}{12}.$$

The first-period utility from the deviation is

$$\hat{U}_{i1}(0.8\alpha \mid 0, 1) = (2\alpha - 1) \left(\frac{0.8\alpha}{2\alpha} - \frac{(0.8\alpha)^2}{4\alpha} \right) = (2\alpha - 1)(0.4 - 0.16\alpha).$$

Since

$$\beta_{i1}^{-1}(0.8\alpha) = 0.8,$$

the outsider interprets the deviation as if the insider were of type $w_{i1} = 0.8$. Therefore

$$L(0.8) = 0.8, \quad M = \frac{0.8}{2} = 0.4,$$

and the continuation utility is

$$\hat{U}_{i2}(0.8\alpha \mid 0, 1) = \frac{1}{6} - \frac{0.8^3}{48} + \frac{0.8^2}{8} + \frac{2-0.8}{4}(1-0.4)^2 + \frac{1}{2}(1-0.4) = 0.644\bar{6}.$$

Hence the deviation utility is

$$\hat{U}_i(0.8\alpha \mid 0, 1) = (2\alpha - 1)(0.4 - 0.16\alpha) + 0.644\bar{6}.$$

Then let's look at the difference between two utility values:

$$\Delta(\alpha) := \hat{U}_i(0.8\alpha \mid 0, 1) - U_i^{\text{cand}}(\alpha \mid 0, 1).$$

Then

$$\Delta(\alpha) = 0.18\alpha^2 - 0.29\alpha + \frac{121}{750}.$$

This expression is strictly positive for all $\alpha \in (\frac{1}{2}, 1]$. Therefore

$$\hat{U}_i(0.8\alpha \mid 0, 1) > U_i^{\text{cand}}(\alpha \mid 0, 1) \quad \text{for all } \alpha \in \left(\frac{1}{2}, 1\right].$$

Hence every linear strategy $\beta_{i1}(w_{i1}) = \alpha w_{i1}$ admits a profitable downward deviation for the type $(v_{i1}, c) = (0, 1)$, and is therefore not sequentially rational.

Proof of Lemma 2

Suppose the insider's first-period strategy used for beliefs is truthful:

$$\beta_{i1}(c, v) = c + v.$$

Then, after observing bid b , the outsider's posterior is

$$c \mid b \sim U[A(b), D(b)],$$

where

$$A(b) = \max\{0, b - 1\}, \quad D(b) = \min\{1, b\}.$$

In period 1, the outsider's truthful-benchmark best response is

$$\beta_{o1}(v_{o1}) = 2v_{o1}.$$

Therefore, if an insider of type (c, v) bids $b \in [0, 2]$, his current expected payoff is

$$U_{i1}(b; c, v) = \int_0^{b/2} (c + v - 2v_{o1}) dv_{o1} = \frac{(c + v)b}{2} - \frac{b^2}{4}.$$

Now consider the continuation payoff

$$U_{i2}(c \mid b) = \int_0^1 \int_0^1 \max\{v_i + c - \beta_{o2}(v_{o2} \mid b), 0\} dv_i dv_{o2}.$$

For $b \in [0, 1]$, the truthful posterior is

$$c \mid b \sim U[0, b].$$

The outsider's second-period strategy is

$$\beta_{o2}(v_{o2} \mid b) = \begin{cases} 2v_{o2}, & 0 \leq v_{o2} \leq b/2, \\ v_{o2} + b/2, & b/2 \leq v_{o2} \leq 1 - b/2, \\ b - 1 + 2v_{o2}, & 1 - b/2 \leq v_{o2} \leq 1. \end{cases}$$

$\beta_{o2}(v_{o2} \mid b)$ constitutes a strictly increasing transformation of the random variable v_{o2} , thus the distribution of the outsider's second-period bid $S = \beta_{o2}(v_{o2} \mid b)$ has density

$$f_S(s) = \begin{cases} 1/2, & s \in [0, b], \\ 1, & s \in [b, 1], \\ 1/2, & s \in [1, 1 + b]. \end{cases}$$

For a given outsider bid s , define

$$\phi(s, c) = \int_0^1 \max\{v_{i2} + c - s, 0\} dv_{i2}.$$

Then

$$\phi(s, c) = \begin{cases} \frac{1}{2} + c - s, & s \leq c, \\ \frac{1}{2}(1 + c - s)^2, & c < s < c + 1, \\ 0, & s \geq c + 1. \end{cases}$$

Therefore,

$$U_{i2}(c \mid b) = \int \phi(s, c) f_S(s) ds.$$

We only need the derivative with respect to b . There are two relevant cases.

Case 1: $0 \leq b \leq c$

A direct calculation gives

$$\frac{\partial U_{i2}(c | b)}{\partial b} = \frac{b^2 - 2bc + 2b + c^2 - 2c - 1}{4}.$$

The current payoff derivative is

$$\frac{\partial U_{i1}(b; c, v)}{\partial b} = \frac{c + v - b}{2}.$$

Hence the total derivative is

$$\frac{\partial}{\partial b} [U_{i1}(b; c, v) + U_{i2}(c | b)] = \frac{(b - c)^2 + 2v - 1}{4}.$$

So an interior optimum in the region $b \leq c$ solves

$$(b - c)^2 + 2v - 1 = 0.$$

Thus

$$b = c - \sqrt{1 - 2v}.$$

This solution is valid only if $v < 1/2$ and $b \geq 0$, i.e.

$$c - \sqrt{1 - 2v} \geq 0.$$

Equivalently,

$$v \geq \frac{1 - c^2}{2}.$$

Therefore, in this lower region,

$$BR(c, v) = 0 \quad \text{if} \quad v \leq \frac{1 - c^2}{2},$$

and otherwise

$$BR(c, v) = c - \sqrt{1 - 2v}.$$

Case 2: $c \leq b \leq 1$

A direct calculation gives

$$\frac{\partial U_{i2}(c | b)}{\partial b} = -\frac{(b - c - 1)^2}{4}.$$

Therefore,

$$\frac{\partial}{\partial b} [U_{i1}(b; c, v) + U_{i2}(c | b)] = \frac{2v - 1 - (b - c)^2}{4}.$$

An interior optimum in this region solves

$$2v - 1 - (b - c)^2 = 0.$$

Thus

$$b = c + \sqrt{2v - 1}.$$

This solution is valid only if $v \geq 1/2$ and $b \leq 1$, i.e.

$$c + \sqrt{2v - 1} \leq 1.$$

Equivalently,

$$v \leq \frac{1 + (1 - c)^2}{2}.$$

Therefore, in this upper region,

$$BR(c, v) = c + \sqrt{2v - 1}$$

when

$$\frac{1}{2} \leq v < \frac{1 + (1 - c)^2}{2},$$

and

$$BR(c, v) = 1 \quad \text{if} \quad v \geq \frac{1 + (1 - c)^2}{2}.$$

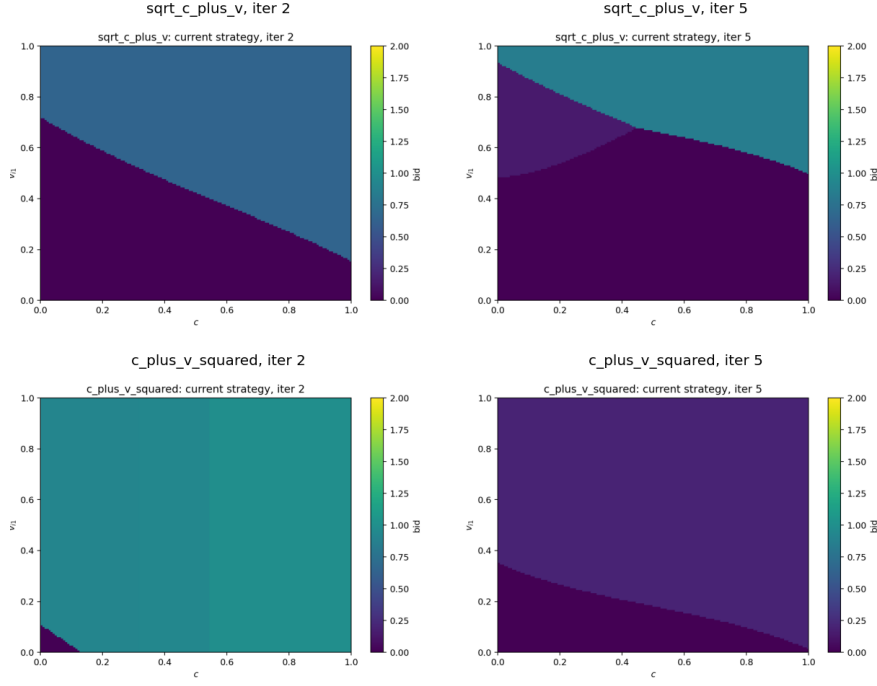


Figure 12: Best responses iterations 2, 5 starting from the strategy $\beta_{i1}(c, v_{i1}) = \sqrt{c} + v_{i1}$ and $\beta_{i1}(c, v_{i1}) = c + v_{i1}^2$

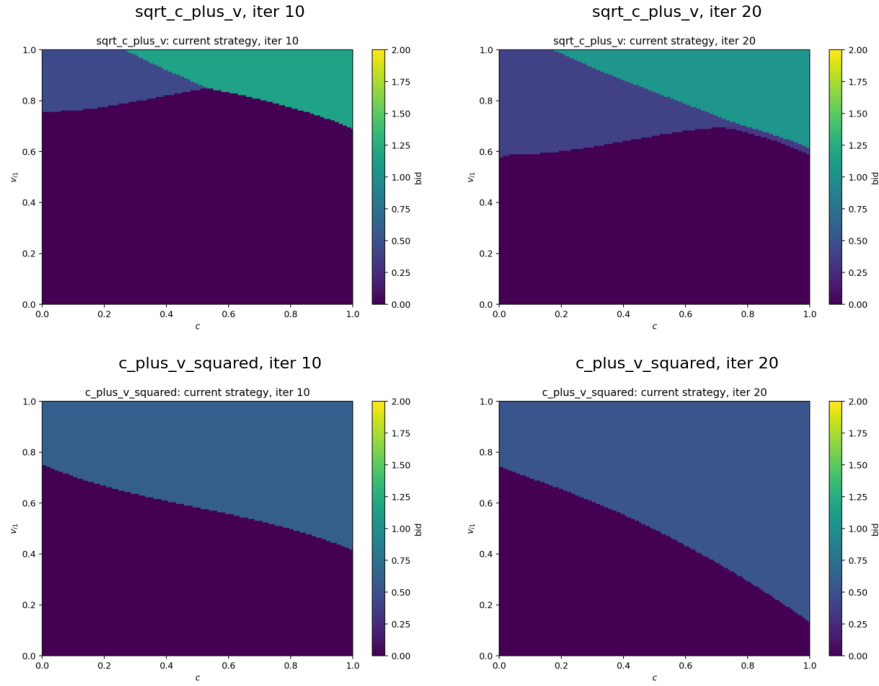


Figure 13: Best responses iterations 10, 20 starting from the strategy $\beta_{i1}(c, v_{i1}) = \sqrt{c} + v_{i1}$ and $\beta_{i1}(c, v_{i1}) = c + v_{i1}^2$